

MP363 – Quantum Mechanics I

Problem Set 6

Due by 5pm on Thursday, 8 January 2015

Please note that this Problem Set is completely OPTIONAL. If you choose not to turn it in, it will not harm your continuous assessment mark in the slightest way. However, if you do turn it in, it will be marked and added to your continuous assessment mark (up to a overall maximum of 100%). And since it also covers examinable material, it's to your benefit to at least attempt to do these problems.

1. The bound-state energies of the finite square well are determined by finding the positive values of z satisfying

$$\tan z = \frac{z\sqrt{2v^2 - z^2}}{z^2 - v^2}$$

where $v = \sqrt{ma^2V_0/\hbar^2}$. If these values are $z_1, z_2, \dots, z_N$, then the energies are

$$E_n = \frac{\hbar^2 z_n^2}{2ma^2}, \quad n = 1, \dots, N$$

Argue (graphically or otherwise) that the total number of bound states is

$$N = \left\lfloor \frac{\sqrt{2}v}{\pi} \right\rfloor + 1,$$

where $\lfloor y \rfloor$ (read “the floor of y ”) is the largest integer less than or equal to y .

2. A particle of mass m moves in a finite square well of width a and depth $V_0 = 5\hbar^2/ma^2$.
 - (a) Use the results of the previous problem to determine the number of bound states.
 - (b) Using whatever means you like – graphics, numerics, calculations – find the (approximate) values of the energies of these bound states, expressed as multiples of $\hbar^2/ma^2$.
3.
 - (a) A beam of electrons is directed at a square potential barrier of height $V_0 = 10$ eV. When the beam energy is $E = 5$ eV, the transmission coefficient is found to be $T = 0.01$. From this data, determine the width a of the barrier. (Remember that $m_e = 9.11 \times 10^{-31}$ kg, $\hbar = 1.05 \times 10^{-34}$ J · s and 1 eV = 1.60 × 10⁻¹⁹ J.)
 - (b) The beam energy is then increased to 10 eV; what is the new transmission coefficient?